05506026

Digital Image Processing

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Version 6 July 2025

CONTENTS

INTENSITY TRANSFORMATIONS

Intensity and Spatial

- As we described before, an image consists with spatial and intensity data

$$f(x,y) = \begin{bmatrix} f(0,0) & f(0,1) & f(0,2) \\ f(1,0) & f(1,1) & f(1,2) \; \ldots \\ f(2,0) & f(2,1) & f(2,2) \\ & \vdots & \end{bmatrix}$$

128	30	123
232	123	321
123	77	89
80	255	255

Intensity Transformations

- Intensity transformations are a set of processes to transform the values of image's pixels
- The values BEFORE transformation are denoted by **r**
- The values AFTER transformation are denoted by **s**
- Therefore,

$$s = T(r)$$

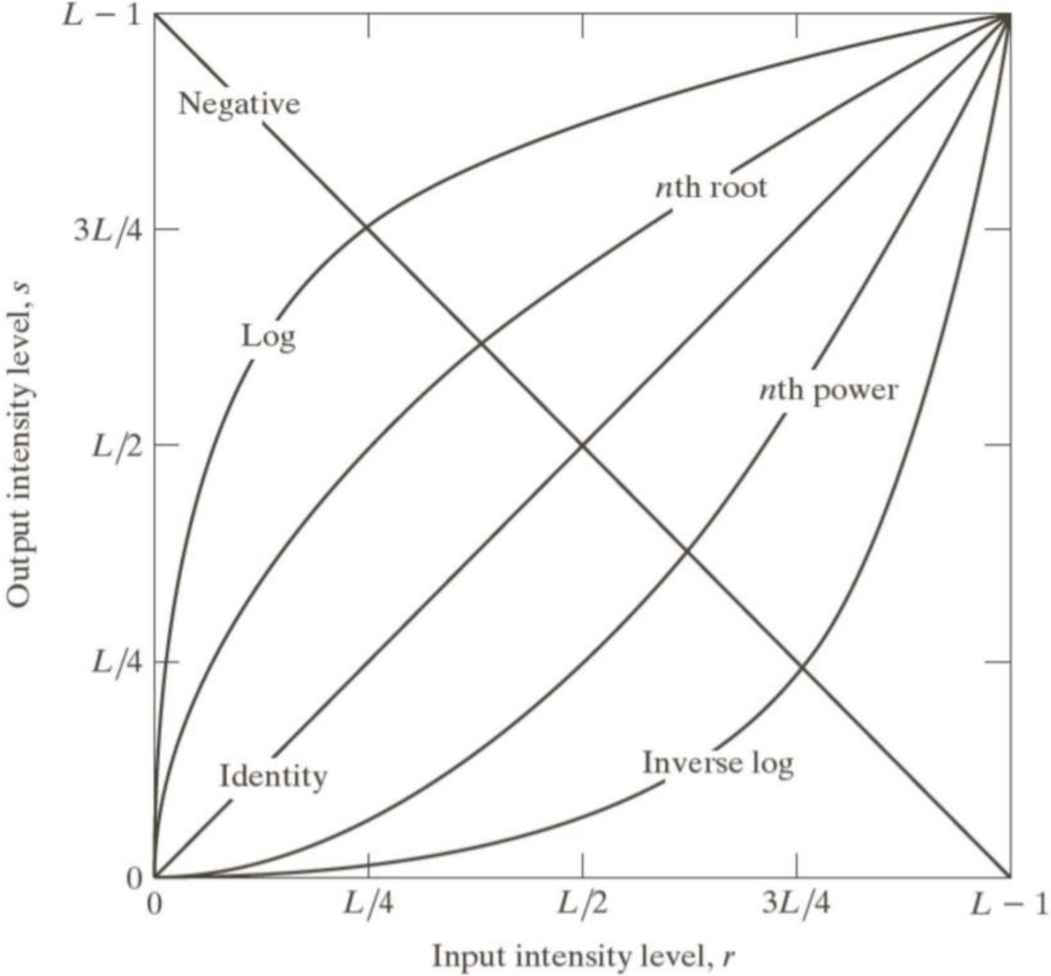


Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Image Negatives

- If the intensity levels are in the range $[0, L-1]$, The negative transformation will be

$$s = L - 1 - r$$

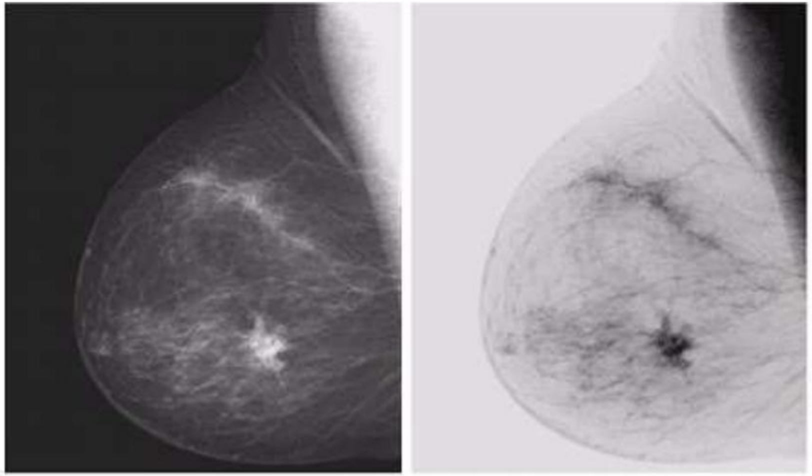


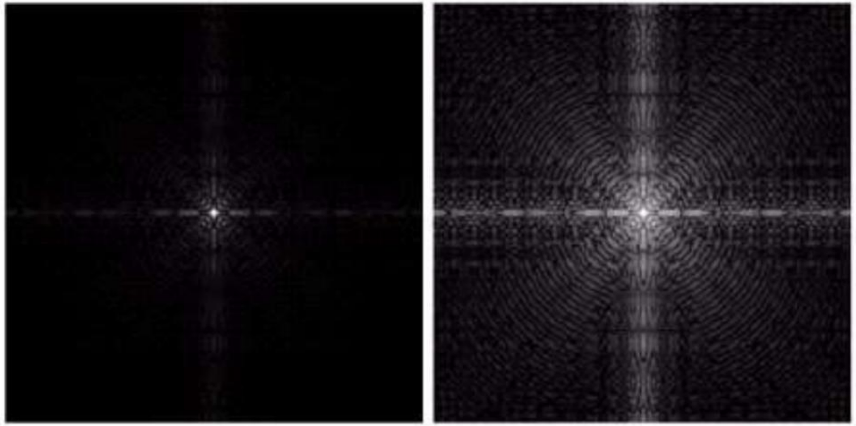
Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Log Transformation

- If the intensity levels are in the range $[0, L-1]$, The log transformation will be

$$s = c\log(1 + r)$$

c is a constant



Original Image **Log Transformed with c=1**

Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Power-Law (Gamma) Transformation

- If the intensity levels are in the range $[0, L-1]$, The power-law transformation will be

$$s = cr^{\gamma}$$

c and γ is a positive constant

Power-Law (Gamma) Transformation

Original Image

Gamma Transformed with c = 1 and γ = 0.6

Gamma Transformed with c = 1 and γ = 0.4

Gamma Transformed with c = 1 and γ = 0.3

Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Power-Law (Gamma) Transformation

Original Image

Gamma Transformed with c = 1 and γ = 3.0

Gamma Transformed with c = 1 and γ = 4.0

Gamma Transformed with c = 1 and γ = 5.0

Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Piecewise-Linear Transformation Functions

- Some images can only be processed and enhanced depending on the piece of image
- Therefore, they need some complex functions

Piecewise-Linear Transformation Functions

Contrast Stretching

- Used to expand the range of intensity levels in an image

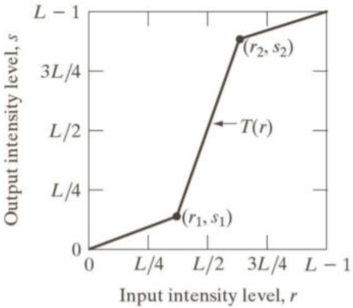


Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Piecewise-Linear Transformation Functions

Intensity-level slicing

- Used to enhance the range of intensity levels of interest
- Normally, there are two ways, to reduce the other intensity out of interest OR highlight the intensity of interest

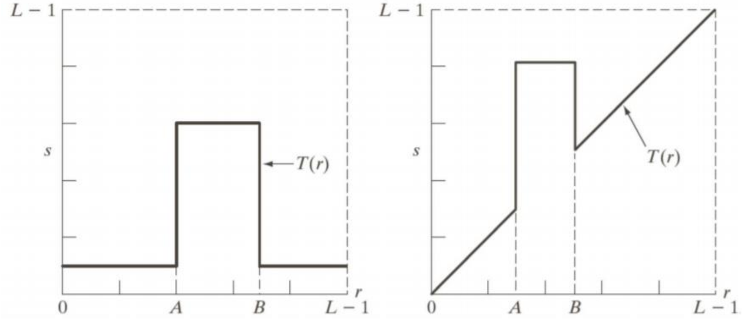


Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Piecewise-Linear Transformation Functions

Original Image

Intensity-level sliced by reducing the other intensity

Intensity-level sliced by highlighting the intensity of interest

Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Piecewise-Linear Transformation Functions

Bit-plane slicing

- Highlight the contributions of specific bits instead of an intensity range

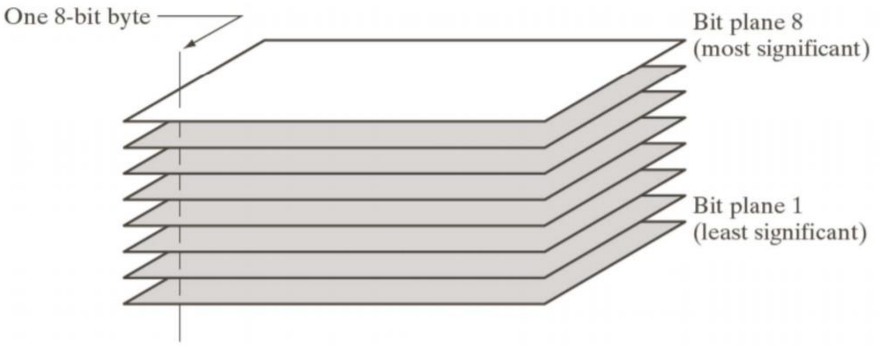


Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Piecewise-Linear Transformation Functions

Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

COLOUR BALANCE

Colour Balance

- Colour Balance, sometimes called White Balance, is the adjustment of the intensities of the colour in every pixels in a digital image.
- Adjusting colour balance affects the temperature of the image

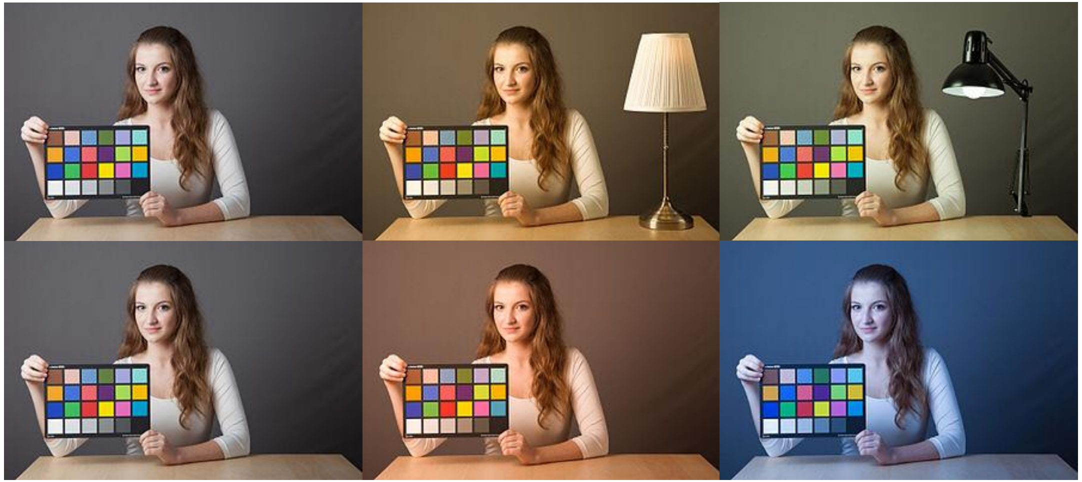


Image from: https://en.wikipedia.org/wiki/Color_balance

Colour Balance

- We can adjust the colour balance using several methods, but most of them are to multiply each colour channel with normalised values

$$\begin{bmatrix} R' \\ G' \\ B' \end{bmatrix} = \begin{bmatrix} R'_w/255 & 0 & 0 \\ 0 & G'_w/255 & 0 \\ 0 & 0 & B'_w/255 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

Image from: https://en.wikipedia.org/wiki/Color_balance

Colour Temperature

- Colour Temperature is a temperature of an ideal black-body radiator of a light source
- In practice, the light source is corresponding to the radiation of black body, which is ranged from

red → orange → yellow → white → blue

- Colour temperature is normally expressed in kelvins or K

Temperature	Source
1700 K	Match flame, low pressure sodium lamps (LPS/SOX)
1850 K	Candle flame, sunset/sunrise
2400 K	Standard incandescent lamps
2550 K	Soft white incandescent lamps
2700 K	"Soft white" compact fluorescent and LED lamps
3000 K	Warm white compact fluorescent and LED lamps
3200 K	Studio lamps, photofloods, etc.
3350 K	Studio "CP" light
5000 K	Horizon daylight
5000 K	Tubular fluorescent lamps or cool white/daylight compact fluorescent lamps (CFL)
5500 – 6000 K	Vertical daylight, electronic flash
6200 K	Xenon short-arc lamp[2]
6500 K	Daylight, overcast
6500 – 9500 K	LCD or CRT screen
15,000 – 27,000 K	Clear blue poleward sky
These temperatures are merely characteristic; there may be considerable variation	

- The RGB values of colour temperatures are as follow

1000 K	#ff3800
1200 K	#ff5300
1400 K	#ff6500
1600 K	#ff7300
1800 K	#ff7e00
2000 K	#ff8912
2200 K	#ff932c
2400 K	#ff9d3f
2600 K	#ffa54f
2800 K	#ffad5e
3000 K	#ffb46b
3200 K	#ffbb78
3400 K	#ffc184
3600 K	#ffc78f
3800 K	#ffcc99
4000 K	#ffd1a3
4200 K	#ffd5ad
4400 K	#ffd9b6
4600 K	#ffddbe
4800 K	#ffe1c6
5000 K	#ffe4ce
5200 K	#ffe8d5
5400 K	#ffebdc
5600 K	#ffeee3
5800 K	#fff0e9
6000 K	#fff3ef
6200 K	#fff5f5
6400 K	#fff8fb
6600 K	#fef9ff
6800 K	#f9f6ff
7000 K	#f5f3ff
7200 K	#f0f1ff
7400 K	#edefff
7600 K	#e9edff
7800 K	#e6ebff

8000 K	#e3e9ff
8200 K	#e0e7ff
8400 K	#dde6ff
8600 K	#dae4ff
8800 K	#d8e3ff
9000 K	#d6e1ff
9200 K	#d3e0ff
9400 K	#d1dfff
9600 K	#cfddff
9800 K	#cedcff
10000 K	#ccdbff
10200 K	#cadaff
10400 K	#c9d9ff
10600 K	#c7d8ff
10800 K	#c6d8ff
11000 K	#c4d7ff
11200 K	#c3d6ff
11400 K	#c2d5ff
11600 K	#c1d4ff
11800 K	#c0d4ff
12000 K	#bfd3ff
12200 K	#bed2ff
12400 K	#bdd2ff
12600 K	#bcd1ff
12800 K	#bbd1ff
13000 K	#bad0ff
13200 K	#b9d0ff
13400 K	#b8cfff
13600 K	#b7cfff
13800 K	#b7ceff
14000 K	#b6ceff
14200 K	#b5cdff
14400 K	#b5cdff
14600 K	#b4ccff
14800 K	#b3ccff
15000 K	#b3ccff
15200 K	#b2cbff
15400 K	#b2cbff
15600 K	#b1caff
15800 K	#b1caff

16000 K	#b0caff
16200 K	#afc9ff
16400 K	#afc9ff
16600 K	#afc9ff
16800 K	#aec9ff
17000 K	#aec8ff
17200 K	#adc8ff
17400 K	#adc8ff
17600 K	#acc7ff
17800 K	#acc7ff
18000 K	#acc7ff
18200 K	#abc7ff
18400 K	#abc6ff
18600 K	#aac6ff
18800 K	#aac6ff
19000 K	#aac6ff
19200 K	#a9c6ff
19400 K	#a9c5ff
19600 K	#a9c5ff
19800 K	#a9c5ff
20000 K	#a8c5ff
20200 K	#a8c5ff
20400 K	#a8c4ff
20600 K	#a7c4ff
20800 K	#a7c4ff
21000 K	#a7c4ff
21200 K	#a7c4ff
21400 K	#a6c3ff
21600 K	#a6c3ff
21800 K	#a6c3ff
22000 K	#a6c3ff
22200 K	#a5c3ff
22400 K	#a5c3ff
22600 K	#a5c3ff
22800 K	#a5c2ff
23000 K	#a4c2ff
23200 K	#a4c2ff
23400 K	#a4c2ff
23600 K	#a4c2ff
23800 K	#a4c2ff

24000 K	#a3c2ff
24200 K	#a3c1ff
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25000 K	#a3c1ff
25200 K	#a2c1ff
25400 K	#a2c1ff
25600 K	#a2c1ff
25800 K	#a2c1ff
26000 K	#a2c0ff
26200 K	#a2c0ff
26400 K	#a1c0ff
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27000 K	#a1c0ff
27200 K	#a1c0ff
27400 K	#a1c0ff
27600 K	#a1c0ff
27800 K	#a0c0ff
28000 K	#a0bfff
28200 K	#a0bfff
28400 K	#a0bfff
28600 K	#a0bfff
28800 K	#a0bfff
29000 K	#a0bfff
29200 K	#a0bfff
29400 K	#9fbfff
29600 K	#9fbfff
29800 K	#9fbfff

Full RGB values can be obtained at

http://www.vendian.org/mncharity/dir3/blackbody/UnstableURLs/bbr_color.html

HISTOGRAM

Histogram

- Histogram is a representation of the distribution of intensity data in a digital image
- The intensity data (or so called “tonal data”) in a range of values will be “bucketed” to create a graph
- Horizontal axis of the graph represents the variations of tonal data, while the vertical axis represents the number of pixels in that tone.

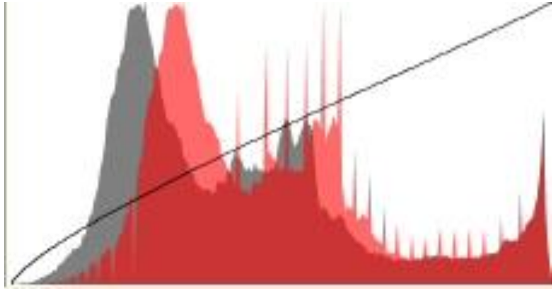
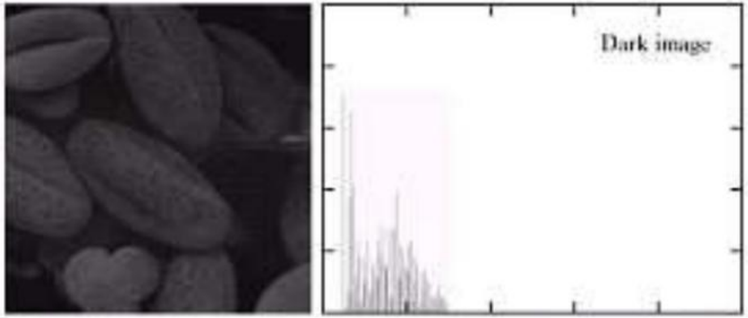


Image from: https://en.wikipedia.org/wiki/Image_histogram

Histogram

- Histogram can be used to show the characteristics of a digital image in a glance



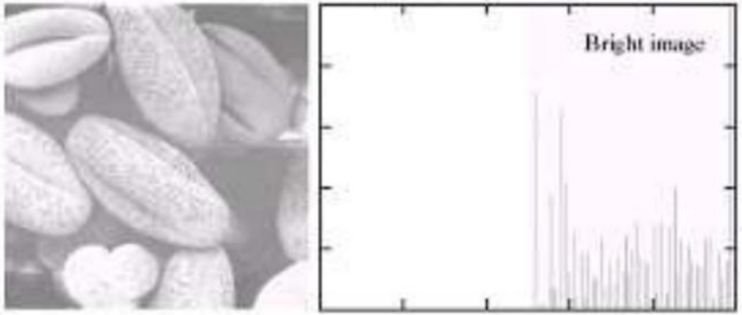


Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Histogram

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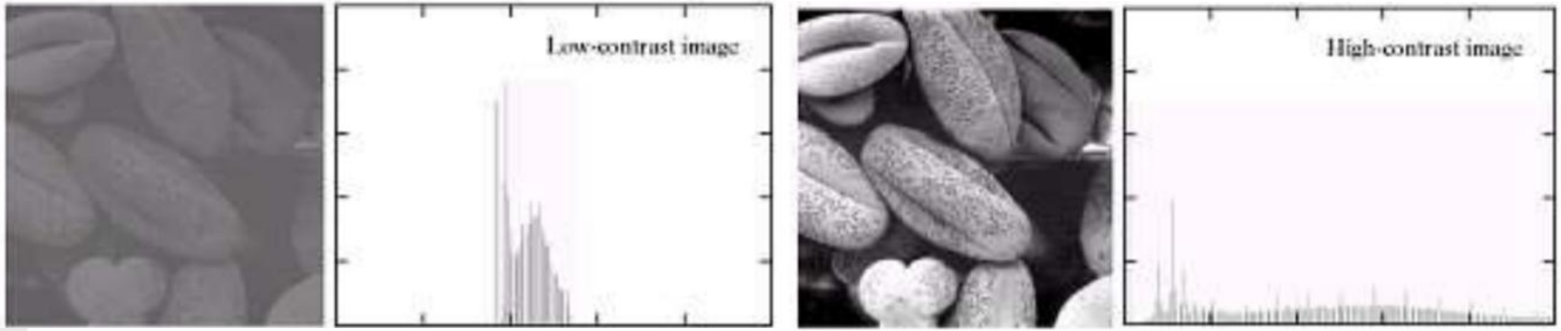


Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Histogram

- Histogram can also be used for thresholding (or binarization) method, Edge detection, Image segmentation, etc.

Image from: https://en.wikipedia.org/wiki/Thresholding_(image_processing)

Histogram

- Histogram can also be shown in colour model which each colour channel will be graphed individually

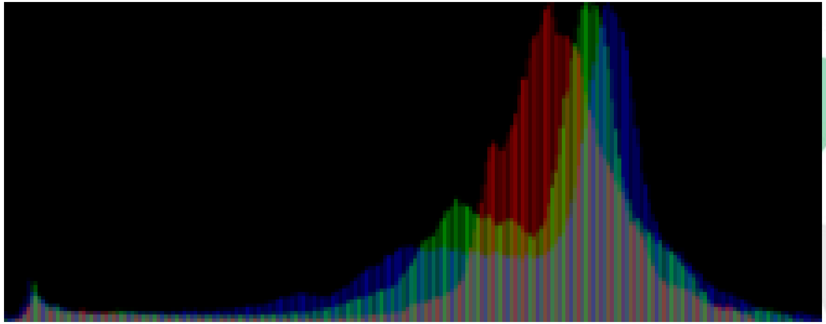


Image from: https://en.wikipedia.org/wiki/Color_histogram

Histogram

- Histogram of a digital image with intensity levels in the range $[0, \mathrm{L}-1]$ can be represented with a discrete function

$$h(r_k) = n_k$$

where

k is in the range $[0, \mathrm{L}-1]$

r_k is the *k*th intensity value

n_k is the number of pixels with intensity r_k

Histogram

- It is common practice to normalise a histogram as

$$p(r_k) = \frac{n_k}{MN}$$

where

k is in the range $[0, \mathrm{L} - 1]$

r_k is the kth intensity value

n_k is the number of pixels with intensity r_k

M is the row dimension of the image, N is the column dimension of the image

Histogram

We can call this normalised histogram as probability of occurrence of intensity level in an image. Therefore,

$$\sum p(r_k) = 1$$

CONTRAST ENHANCEMENT

Contrast

- Contrast is the difference in intensity values of a digital image

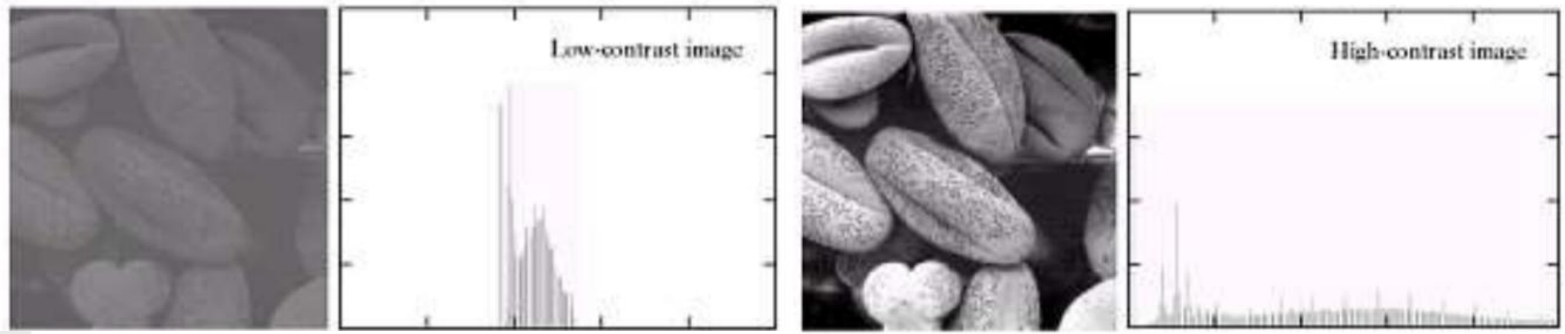
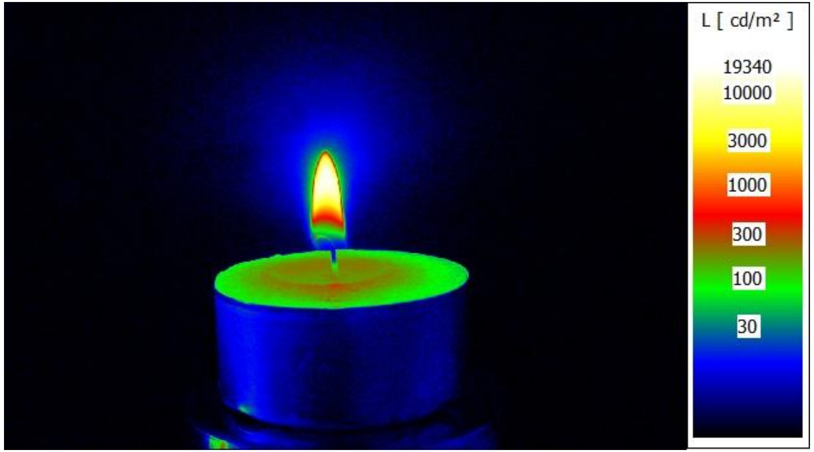


Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Contrast

- The contrast of a digital image can be calculated by using luminance value of pixels
- Each RGB colour channel has different luminance value because different emissions from particalur angles



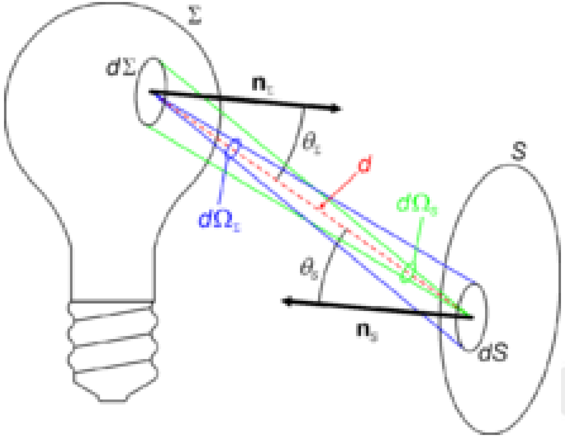


Image from: https://en.wikipedia.org/wiki/Luminance

Contrast

- Converting an RGB image to a luminance image has several standards. However, a common strategy is using CIE 1931 XYZ Colour Space to calculate as follow.

$$Y_{\text{linear}} = 0.2126R_{\text{linear}} + 0.7152G_{\text{linear}} + 0.0722B_{\text{linear}}$$

Contrast

- Once the RGB image is converted to luminance image, the contrast of the image can be calculated by the standard deviation of the pixel intensities, this method is called RMS Contrast

$$\sqrt{\frac{1}{MN}\sum_{i=0}^{N-1}\sum_{j=0}^{M-1}(I_{ij}-\bar{I})^2}$$

Contrast Enhancement

- Contrast of an image can be enhanced by several methods. In this class, we will talk about 2 methods
  - Normalisation
  - Histogram equalisation

Normalisation

- Normalisation, often called Contrast Stretching, is a process that change the range of pixel intensity values

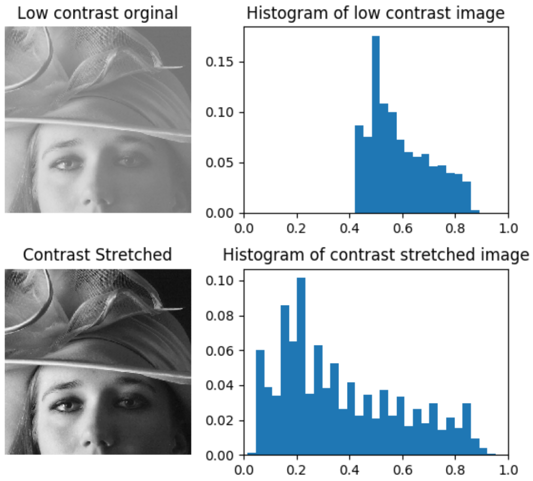


Image from: https://staff.fnwi.uva.nl/r.vandenboomgaard/IPCV20162017/_images/contraststretch.png

Normalisation

- Normalisation can be processed by using the formula

$$I_N = (I - \mathrm{Min})\frac{\mathrm{newMax} - \mathrm{newMin}}{\mathrm{Max} - \mathrm{Min}} + \mathrm{newMin}$$

Histogram equalisation

- Histogram equalisation is a method to adjust the contrast of an image by distribute the pixel intensities equally

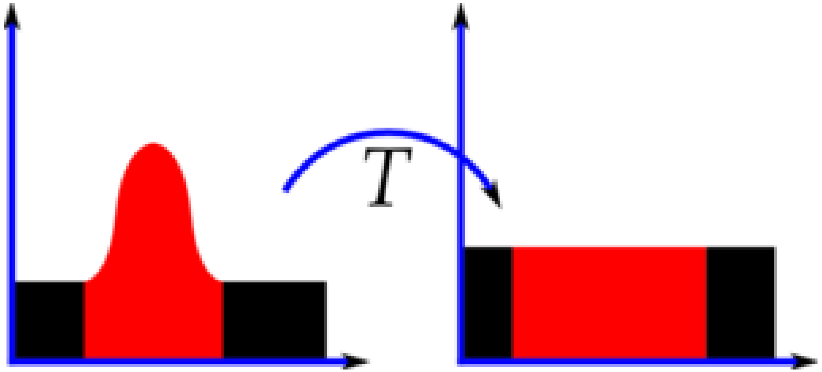


Image from: https://en.wikipedia.org/wiki/Histogram_equalization

Histogram equalisation

- Histogram equalisation can be processed by converting histogram into the cumulative distribution function (CDF)

Value	Count	Value	Count	Value	Count	Value	Count	Value	Count
52	1	64	2	72	1	85	2	113	1
55	3	65	3	73	2	87	1	122	1
58	2	66	2	75	1	88	1	126	1
59	3	67	1	76	1	90	1	144	1
60	1	68	5	77	1	94	1	154	1
61	4	69	3	78	1	104	2		
62	1	70	4	79	2	106	1		
63	2	71	2	83	1	109	1		

v, Pixel Intensity	cdf(v)
52	1
55	4
58	6
59	9
60	10
61	14
62	15
63	17
64	19
65	22
66	24
67	25
68	30
69	33
70	37
71	39
72	40
73	42
75	43
76	44

Histogram equalisation

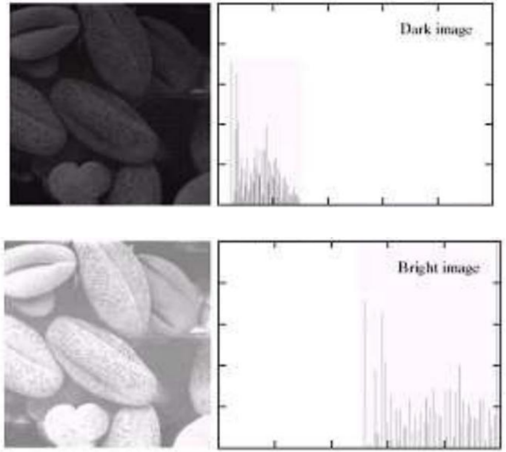
- After that, the value of each pixel will be equalised using the following formula

$$h(v) = \mathrm{round}\left(\frac{cdf(v) - cdf_{min}}{(M \times N) - cdf_{min}} \times (L - 1)\right)$$

- The value will be calculated by stretching the range of histogram to fill the whole range of possible pixel intensities

Histogram equalisation

- The image that is performed histogram equalisation will have better distribution of pixel intensities as example below



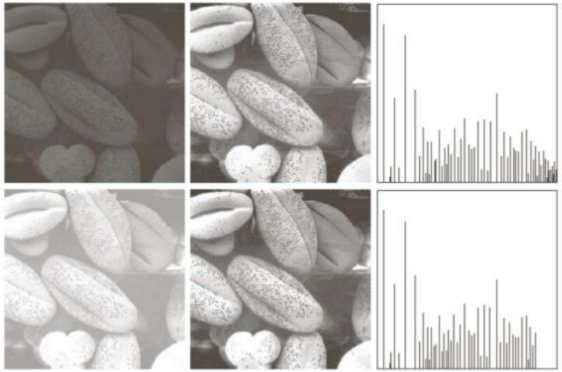


Image from: Gonzalez, R., & Woods, R. (2014). Digital image pro cessing (3rd ed.). New Delhi: Dorling Kindersley.

Histogram equalisation

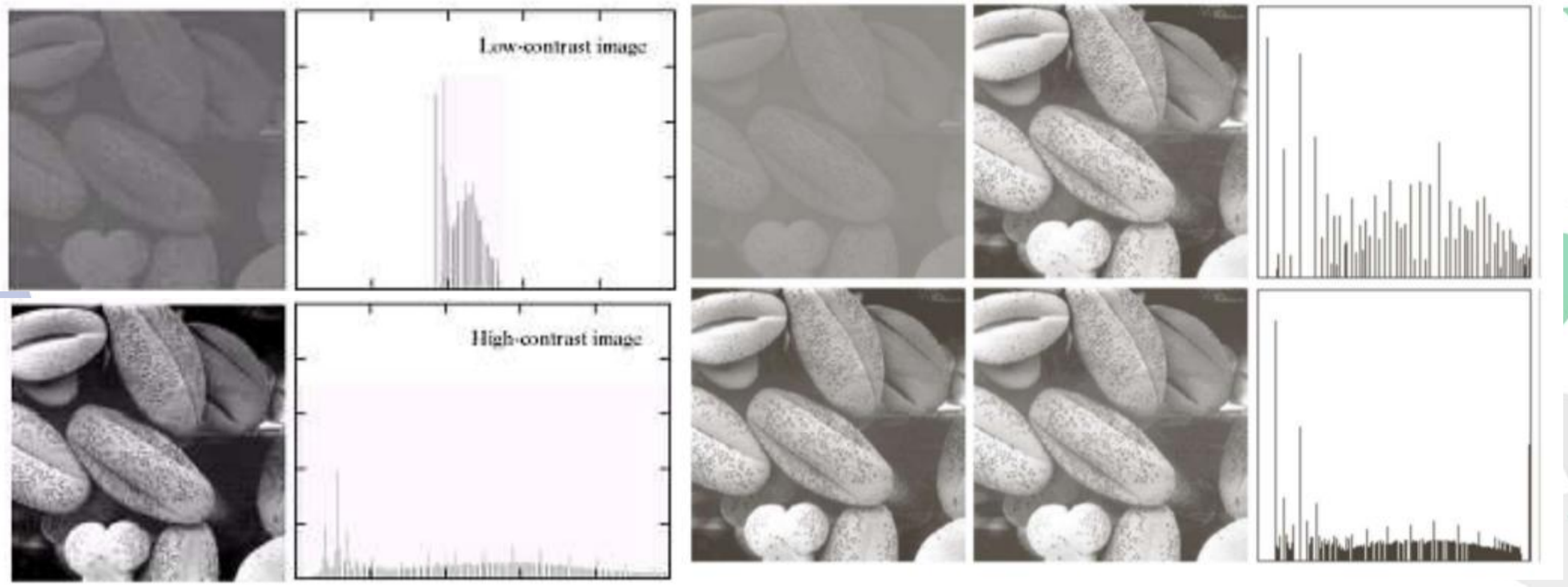
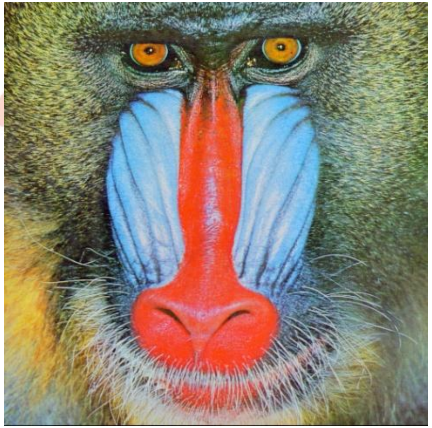


Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Histogram equalisation

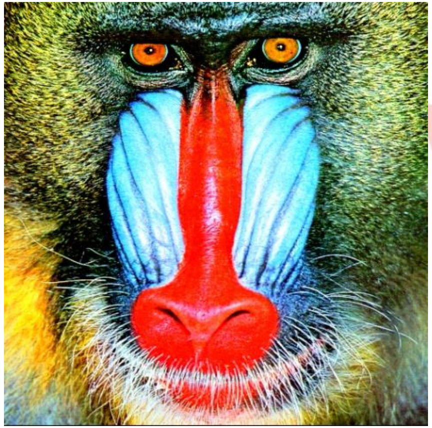
- However, there will be some problems with colour image. Histogram equalisation can improve the contrast of the image, but the colour temperature will be changed.
- This is one of the known research topics of digital image processing

Histogram equalisation



Original Image

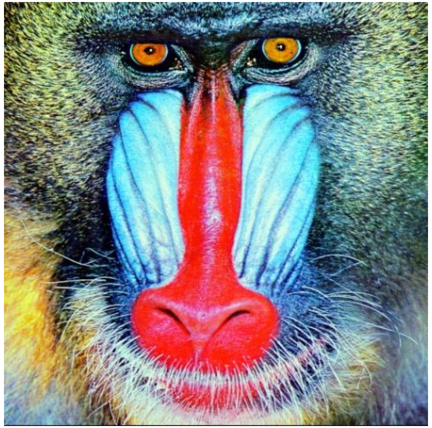
Low Contrast Image



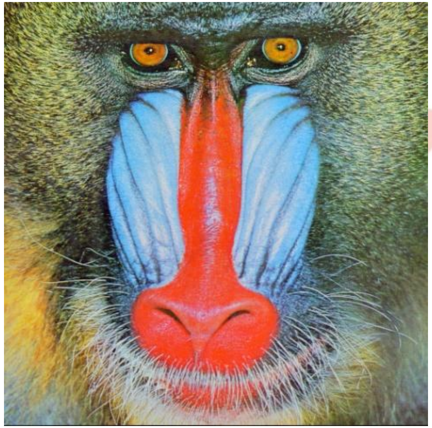
High Contrast Image

Histogram equalisation

Low Contrast Image

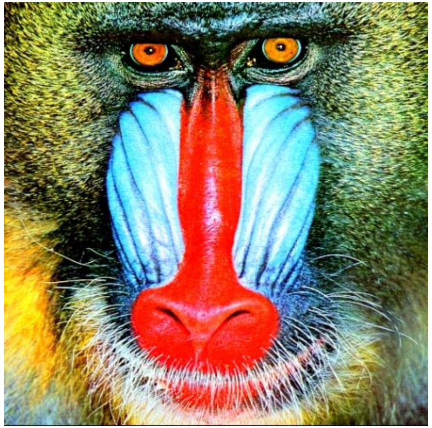


Equalised Image

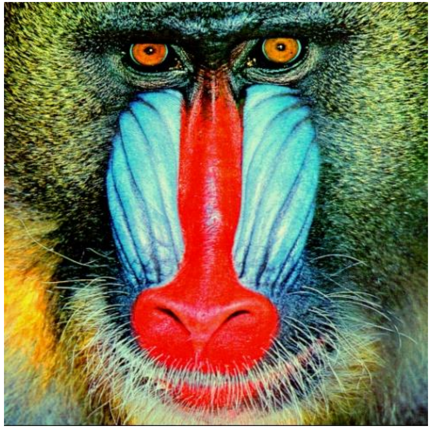


Original Image

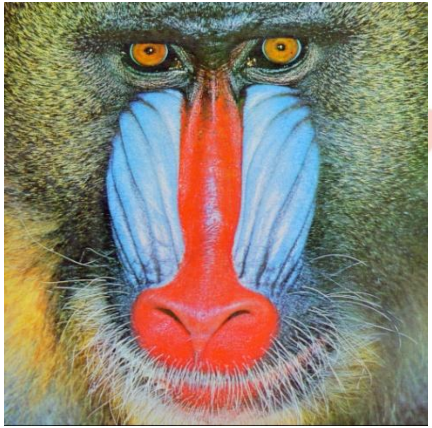
Histogram equalisation



High Contrast Image



Equalised Image



Original Image

End of Week 3